In [1]:
```
import pandas as pd
import os
print(os.getcwd())
print(os.listdir())
```

```
C:\Users\hp\Customer_behaviour
['.ipynb_checkpoints', 'Customer_behaviour.ipynb']
```

In [2]:
```
df = pd.read_csv(r"D:\customer_shopping_behavior.csv")
```

In [4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   str    
 3   Item Purchased          3900 non-null   str    
 4   Category                3900 non-null   str    
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   str    
 7   Size                    3900 non-null   str    
 8   Color                   3900 non-null   str    
 9   Season                  3900 non-null   str    
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str    
 12  Shipping Type           3900 non-null   str    
 13  Discount Applied        3900 non-null   str    
 14  Promo Code Used         3900 non-null   str    
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   str    
 17  Frequency of Purchases  3900 non-null   str    
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

In [5]:
```
df.describe(include='all')
```

Out[5]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>Customer ID</th> <th>Age</th> <th>Gender</th> <th>Item Purchased</th> <th>Category</th> <th>Purchase Amount (USD)</th> <th>Location</th> <th>Size</th> <th>Color</th> <th>Season</th> <th>Review Rating</th> <th>Subscription Status</th> <th>Shipping Type</th> <th>Discount Applied</th> <th>Promo Code Used</th> <th>Previous Purchases</th> <th>Payment Method</th> <th>Frequency of Purchases</th> </thead> <tbody> <th>count</th> <th>unique</th> <th>top</th> <th>freq</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50% </th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>

3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	390
NaN	NaN	2	25	4	NaN	50	4	25	
NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring
NaN	NaN	2652	171	1737	NaN	96	1755	177	99
1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN
1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN
1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN
975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN
1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN
2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN
3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [7]:
```
df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x:x.fi
```

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating              0
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [9]:
```
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ','_')
```

df.columns

In [10]:
```
df.columns
```

Out[10]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

In [11]:
```
df=df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
```

In [12]:
```
df.columns
```

Out[12]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

In [13]:
```
#create a column age_group
labels=['Young Adult','Adult','Middle','Senior']
df['age_group']=pd.qcut(df['age'],q=4,labels=labels)
```

In [14]:
```
df[['age','age_group']].head(10)
```

Out[14]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>age</th> <th>age_group</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> </tbody> <tbody></tbody>

55		Middle
19		Young Adult
50		Middle
21		Young Adult
45		Middle
46		Middle
63		Senior
27		Young Adult
26		Young Adult
57		Middle

In [15]:
```
# create column purchase_frequency_days
frequency_mapping = {
    "Fortnightly": 14,
    "Weekly": 7,
```

```
    "Monthly": 30,
    "Quarterly": 90,
    "Bi-Weekly": 14,
    "Annually": 365,
    "Every 3 Months": 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

In [16]:
```
df['purchase_frequency_days']
```

Out[16]:
```
0        14
1        14
2         7
3         7
4       365
       ...
3895      7
3896     14
3897     90
3898      7
3899     90
Name: purchase_frequency_days, Length: 3900, dtype: int64
```

In [17]:
```
df.isnull().sum()
```

Out[17]:
```
customer_id               0
age                       0
gender                    0
item_purchased            0
category                  0
purchase_amount           0
location                  0
size                      0
color                     0
season                    0
review_rating             0
subscription_status       0
shipping_type             0
discount_applied          0
promo_code_used           0
previous_purchases        0
payment_method            0
frequency_of_purchases    0
age_group                 0
purchase_frequency_days   0
dtype: int64
```

In [18]:
```
df
```

Out[18]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customer_id</th> <th>age</th> <th>gender</th> <th>item_purchased</th> <th>category</th> <th>purchase_amount</th> <th>location</th> <th>size</th> <th>color</th> <th>season</th> <th>review_rating</th> <th>subscription_status</th> <th>shipping_type</th> <th>discount_applied</th> <th>promo_code_used</th> <th>previous_purchases</th> <th>payment_method</th> <th>frequency_of_purchases</th> <th>age_group</th> <th>purchase_frequency_days</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>3895</th> <th>3896</th> <th>3897</th> <th>3898</th> <th>3899</th> </tbody> <tbody></tbody>

1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes
...	...	...	...	...	...	...	...	...	...	...	...
3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No
3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No
3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No
3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No
3900	52	Female	Handbag	Accessories	81	California	M	Beige	Spring	3.1	No

3900 rows × 20 columns

In [19]:
```
df.drop('promo_code_used',inplace=True,axis=1)
```

In [20]:
```
df.columns
```

Out[20]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')
```

In [21]:
```
!pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in C:\Users\hp\AppData\Local\Programs
\Python\Python314\Lib\site-packages (2.9.12)
Requirement already satisfied: sqlalchemy in C:\Users\hp\AppData\Local\Programs\Pyth
on\Python314\Lib\site-packages (2.0.52)
Requirement already satisfied: greenlet>=1 in C:\Users\hp\AppData\Local\Programs\Pyt
hon\Python314\Lib\site-packages (from sqlalchemy) (3.5.5)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\hp\AppData\Local
\Programs\Python\Python314\Lib\site-packages (from sqlalchemy) (4.16.0)
[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [22]:
```
from sqlalchemy import create_engine
from urllib.parse import quote_plus

# My postgresql connection
username = "postgres"
from getpass import getpass
password = getpass("Enter your PostgreSQL password: ")
host = "localhost"
port = "5432"
database = "customer_behaviour"

encoded_password = quote_plus(password)

engine = create_engine(
    f"postgresql+psycopg2://{username}:{encoded_password}@{host}:{port}/{database}"
)

# write dataframe to postgresql
table_name = "customer"  # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data is successfully loaded in table '{table_name}' in database '{database}
```

```
Data is successfully loaded in table 'customer' in database 'customer_behaviour'.
```